AWS VPC PEERING PROJECT

Student Project Report

Student Name	Bhushan Chaudhari
Batch	AWS Internship Batch 1
Project Title	AWS VPC Peering Between Two VPCs
Region	To be filled from AWS Console
Completion Date	23 October 2026
Implementation Status	Partial implementation — 2 EC2 instances launched

Prepared honestly from the resources and progress confirmed during the lab. Add your own AWS screenshots before final submission.

1. Project Objective

The objective is to configure two separate Amazon VPC networks and connect them using VPC Peering. The intended design uses a public bastion host in VPC A and private application instances in VPC A and VPC B. Routes and security group rules control communication between the networks.

2. Architecture and CIDR Plan

VPC A — ewb-vpc-a: 10.10.0.0/16
Public subnet: ewb-a-public — 10.10.1.0/24
Private subnet: ewb-a-private — 10.10.2.0/24
VPC B — ewb-vpc-b: 10.20.0.0/16
Private subnet: ewb-b-private — 10.20.1.0/24
Internet gateway: ewb-igw-a, attached to VPC A
Peering connection: ewb-a-b-peer — reported Active by the student.

Logical architecture (planned)

VPC A — 10.10.0.0/16 Public subnet: ewb-a-public Bastion: actual private IP 10.10.1.194 Private subnet: ewb-a-private App-A: actual private IP 10.10.2.139	**VPC Peering** ewb-a-b-peer Status: Active Routes configured on both sides	**VPC B — 10.20.0.0/16** Private subnet: ewb-b-private App-B: not launched yet Reason: AWS account/region resource validation message

Note: This is a logical diagram, not an AWS-generated screenshot. Replace/augment it with actual console screenshots if your trainer requires visual evidence.

3. Resources Created / Configured

Resource	Name / CIDR	Status / Notes
VPC A	ewb-vpc-a / 10.10.0.0/16	Created
VPC B	ewb-vpc-b / 10.20.0.0/16	Created
Public subnet	ewb-a-public / 10.10.1.0/24	Created; associated with public route table
Private subnet A	ewb-a-private / 10.10.2.0/24	Created; associated with private route table
Private subnet B	ewb-b-private / 10.20.1.0/24	Created; associated with B private route table
Internet gateway	ewb-igw-a	Attached to VPC A
Peering	ewb-a-b-peer	Active; routes reported Active
EC2 bastion	ewb-bastion	Running; private IP 10.10.1.194; public IPv4 not confirmed
EC2 App-A	ewb-app-a	Running; private IP 10.10.2.139; no public IP intended
EC2 App-B	ewb-app-b	Not launched; AWS resource validation restriction reported

4. Route Tables and Peering Routes

The student reported that the following route tables have been configured and the peering routes show Active. Record the actual route table IDs and take screenshots from the AWS VPC console before submitting.

Route table	Subnet association	Relevant routes
ewb-rt-a-public	ewb-a-public	10.10.0.0/16 → local 0.0.0.0/0 → ewb-igw-a 10.20.0.0/16 → ewb-a-b-p
ewb-rt-a-private	ewb-a-private	10.10.0.0/16 → local 10.20.0.0/16 → ewb-a-b-peer
ewb-rt-b-private	ewb-b-private	10.20.0.0/16 → local 10.10.0.0/16 → ewb-a-b-peer

5. Security Groups

ewb-sg-bastion: SSH TCP 22 from 106.222.229.130/32 (student-reported public IP).
ewb-sg-app-a: SSH TCP 22 from actual bastion private IP 10.10.1.194/32, as reported by the student. The intended ICMP Echo Request rule for App-B is not verified.
ewb-sg-app-b: Configured in the project plan; App-B is not running because it could not be launched, so end-to-end rules/tests cannot be validated.

6. Verification Results

Test	Result
VPC peering state	Reported Active in AWS Console
Route table entries	Student reported peering routes Active
SSH from laptop to bastion	Not successful; bastion public IPv4 was not visible/confirmed
SSH from bastion to App-A	Not tested; bastion connectivity unavailable
App-A ↔ App-B ping	Not tested; App-B not launched
Controlled ICMP failure and recovery	Not tested

7. Limitation and Next Steps

During this lab, AWS displayed a message that resource access in the selected Region was being validated and additional instances could not be launched until validation completed. As a result, only the bastion and App-A instances were launched. App-B and end-to-end peering connectivity tests remain incomplete. The project is therefore a partial implementation, not a verified end-to-end success.

Once AWS account/Region validation is cleared, launch ewb-app-b in ewb-b-private, confirm its private IP and security group rules, verify both directions of ICMP over the peering connection, test SSH via the bastion, and capture the actual results.

8. What I Learned

This lab helped me practise creating custom VPCs and subnets, attaching an Internet Gateway, associating route tables with subnets, creating and accepting a VPC peering connection, and configuring security groups for restricted SSH and ICMP traffic. I also learned that Active peering alone does not prove connectivity: forward routes, return routes, security group rules, actual host IPs, and successful tests are all required.

9. Screenshots to Add Before Submission

Add your own labelled screenshots if available: (1) VPC A and VPC B with CIDRs; (2) all subnet CIDRs; (3) peering connection showing Active and its ID; (4) all three route tables and subnet associations; (5) security group inbound rules; (6) EC2 instances and their actual IPs; (7) the AWS resource-validation error. Do not include private keys, passwords, access tokens, or key contents.

10. Cleanup

After the trainer has accepted the evidence, remove only lab resources that are no longer needed: terminate the two lab EC2 instances, verify/delete their lab EBS volumes if retained, remove peering routes and the peering connection, delete lab security groups after network interfaces disappear, disassociate/delete lab route tables, delete lab subnets, detach/delete the lab Internet Gateway, and then delete both lab VPCs. Release any Elastic IP only if one was created. Check the Billing console for remaining chargeable resources. Save evidence before cleanup.

Prepared for: AWS Internship Batch 1
Student: Bhushan Chaudhari
Report status: Partial implementation; actual tests are clearly marked as not tested where applicable.